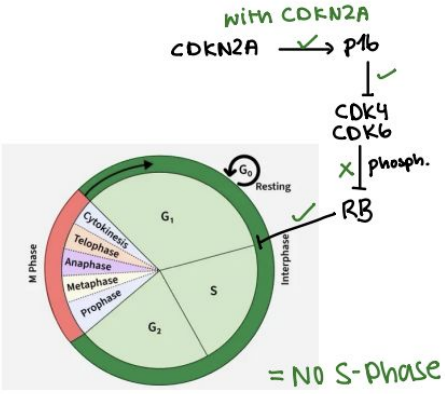
CDKN2A in Gliobastoma - Survival Analysis with Progenetix

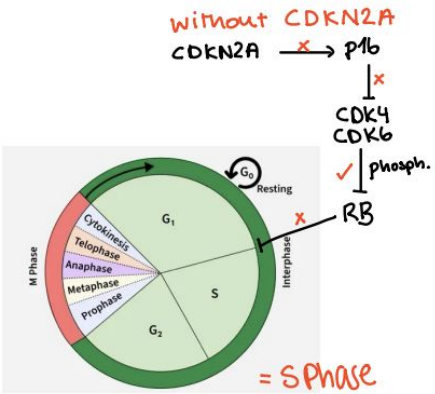
By Siria, Amira and Nami

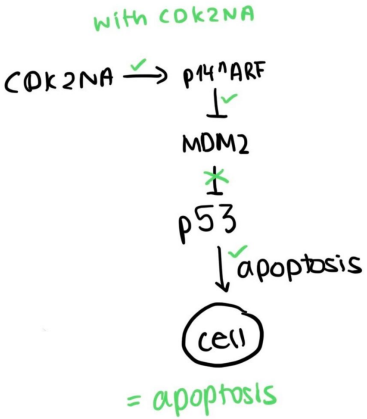
CDKN2A

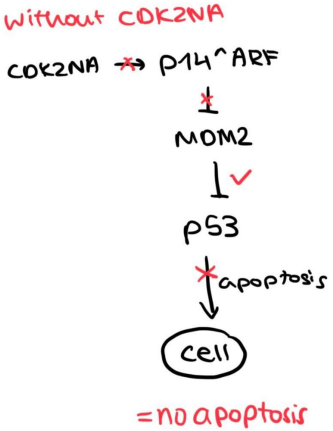
- CDKN2A, chromosome: 9, start: 21967752, end: 21995324
- Cyclin-dependent kinase inhibitor 2A
- encodes two distinct proteins from the same locus
  - p16^INK4a
  - p14^ARF



https://www.geeksforgeeks.org/biology/cell-cycle-and-division/





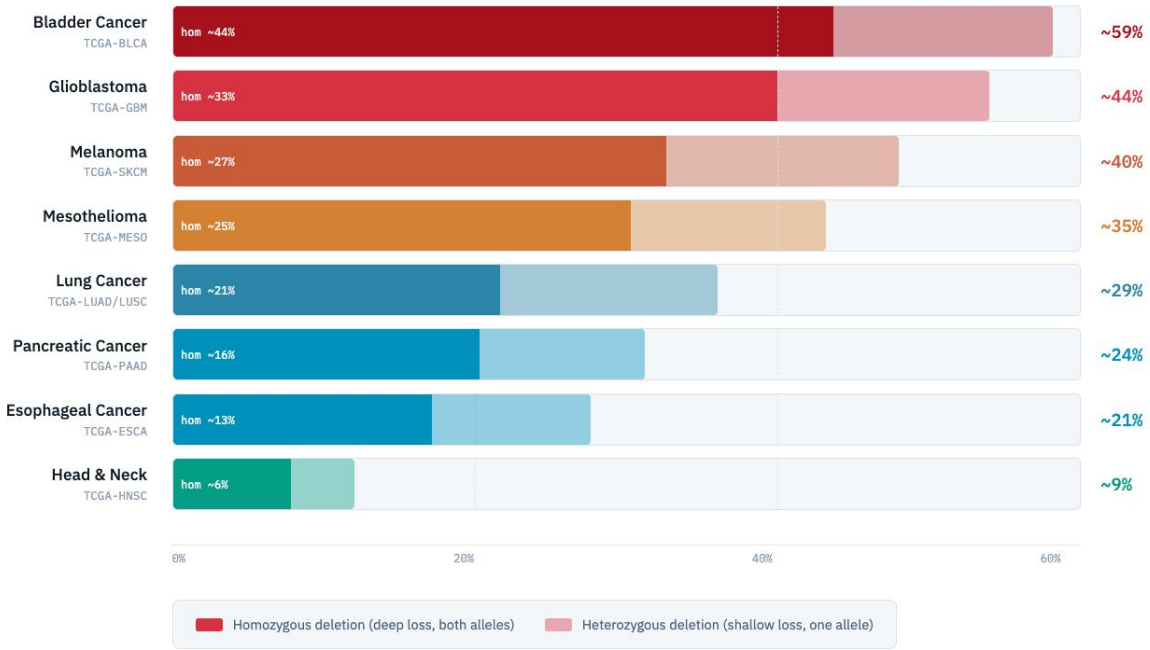


https://www.uniprot.org/uniprotkb/P42771/entry

COPY NUMBER VARIATION · CHR9P21.3

CDKN2A Deletion Frequency Across *Cancer Types*

Homozygous & heterozygous deletion rates by tumour type · Source: TCGA / COSMIC pan-cancer data



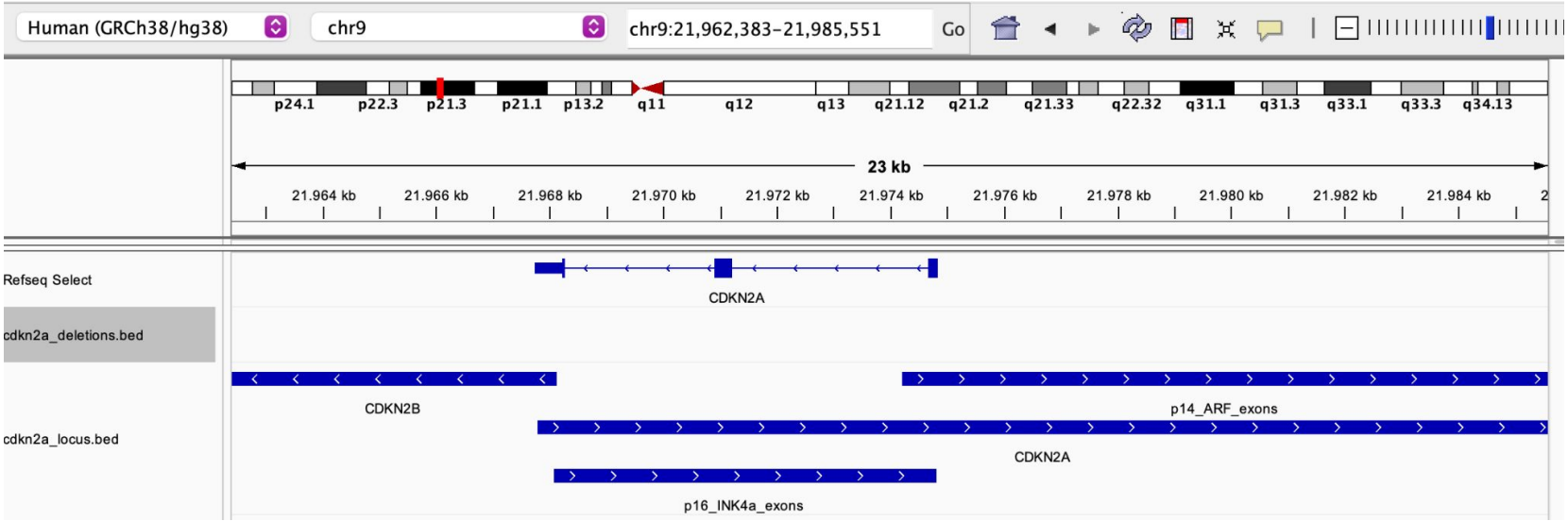
Frequencies are approximate and reflect combined CNV deletion calls (shallow + deep loss) across studies.
Homozygous estimates derived from high-level loss calls; heterozygous from low-level loss calls.

chr9p21.3 · CDKN2A

Langwieder, C. K., Hölzl, D., Hutarew, G., Schlicker, H. U., Alinger-Scharinger, B., Schwartz, C., Sotlar, K., & Kraus, T. F. J. (2026). Loss of CDKN2A/B is a Hallmark of RTK II Glioblastomas. *Journal of Cancer*, *17*(1), 1–9. https://doi.org/10.7150/jca.122609

National Cancer Institute. (2015). *Phase II study of ilorasertib (ABT-348) in patients with CDKN2A deficient solid tumors* (Protocol No. NCT02478320). ClinicalTrials.gov. https://cdn.clinicaltrials.gov/large-docs/20/NCT02478320/Prot_SAP_000.pdf

van de Wetering, M., Francies, H. E., Francis, J. M., Bounova, G., Iorio, F., Pronk, A., van Houdt, W., van Gorp, J., Taylor-Weiner, A., Kester, L., McLaren-Douglas, A., Blokker, J., Jaksani, S., Bartfeld, S., Volckman, R., van Sluis, P., Li, V. S. W., Seepo, S., Sekhar Pedamallu, C., … Clevers, H. (2015). Prospective Derivation of a Living Organoid Biobank of Colorectal Cancer Patients. *Cell*, *161*(4), 933–945. https://doi.org/10.1016/j.cell.2015.03.053



- Deletions at 9p21.3
- Heterozygous deletion: milder phenotype, less aggressive tumors
- Homozygous deletion: zero functional CDKN2A, more aggressive disease, worse survival
- Co-deletion problem: CDKN2B (p15) on 9p21.3 similar job (inhibits CDK4/6)

Workflow & Methods

Data Retrieval

- Sample metadata retrieved from Progenetix via the sampletable API (n = 555, TCGA-GBM)
- Survival data (follow-up time, vital status) fetched separately at individual level via Beacon individuals endpoint
- Merged through individual_id; one biosample retained per individual at random (seed = 42)

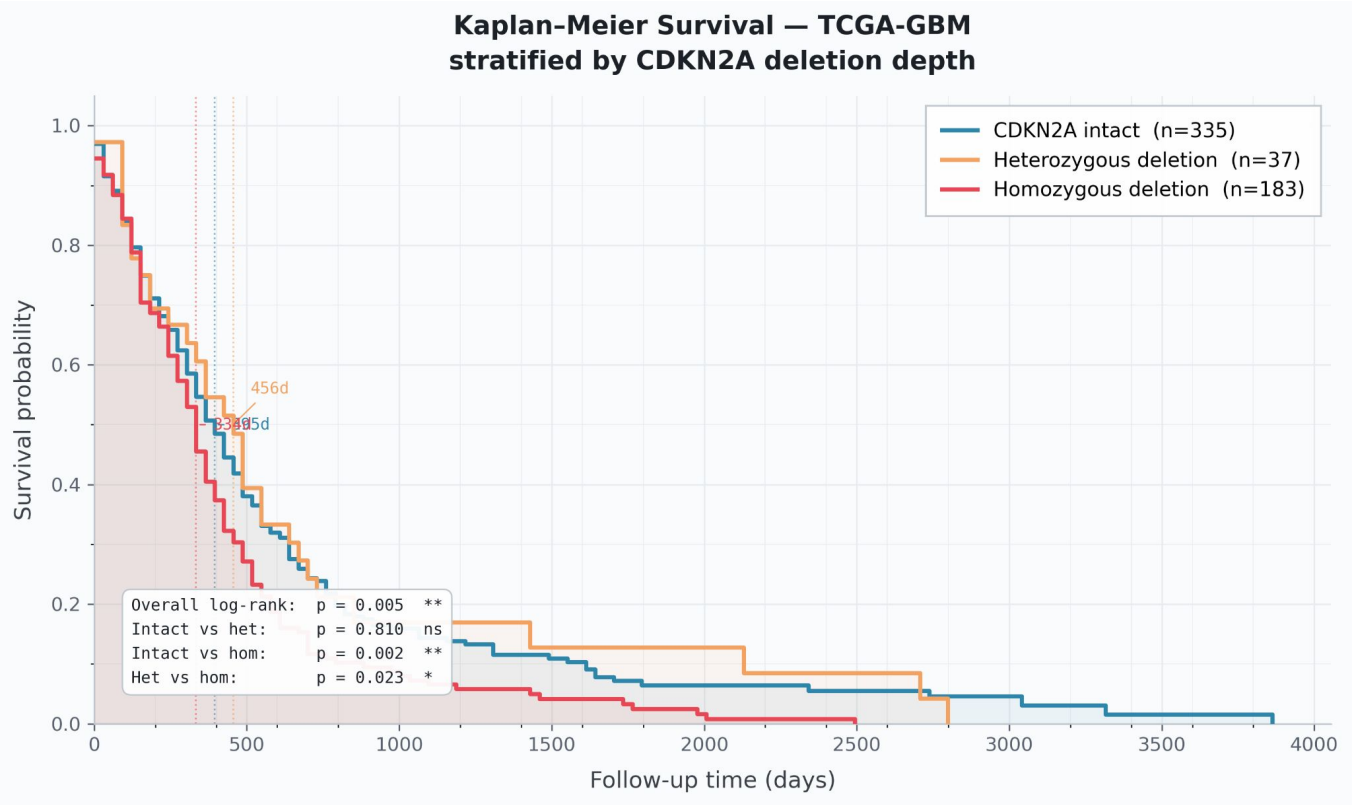
CNV Feature Engineering

- CNV records retrieved per biosample from Progenetix g_variants Beacon endpoint
- Each CNV interval tested for overlap with CDKN2A locus (chr9:21,967,752–21,995,325, GRCh38)
- Deletion depth classified by EFO term: EFO:0030067 → heterozygous, EFO:0020073 → homozygous
- Where both low- and high-level loss were present, homozygous was prioritised

Survival Analysis

- Kaplan–Meier curves estimated for three groups: intact (n = 335), heterozygous (n = 37), homozygous (n = 183)
- Log-rank test applied overall across all three groups and pairwise between each pair
- Cox proportional hazards model fitted on all 555 individuals (no missing covariates)
- Additional covariate: sex (female = 0, male = 1)
- Standard errors via numerical Hessian of the log partial likelihood; p-values from Wald test
- Model discrimination assessed using concordance index (C-index)

Results: Kaplan-Meier Analysis



- Median survival
  - Homozygous loss: 334 days
  - Heterozygous loss: 456 days
  - Intact: 395 days

- CDKN2A loss has a negative effect on long-term survival

Results: Cox Proportional Hazards Model

COX PROPORTIONAL HAZARDS MODEL

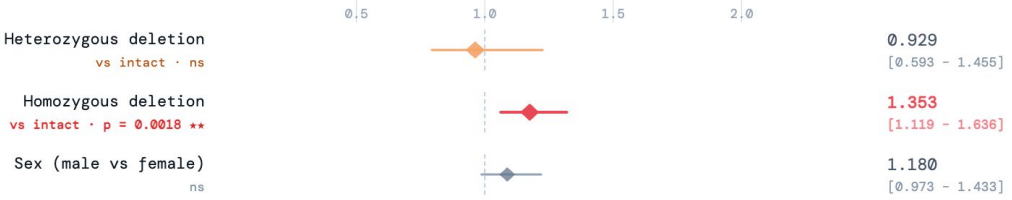
TCGA-GBM

CDKN2A Deletion Depth & Survival

TCGA-GBM · n = 555 · reference category: CDKN2A intact (n = 335) · events = 455

GROUPS ● Intact n = 335 ● Heterozygous n = 37 ● Homozygous n = 183

HAZARD RATIO FOREST PLOT (95% CI) — REFERENCE: INTACT



FEATURE	COEF	SE	Z	P-VALUE	HR	95% CI	SIG.
Heterozygous (vs intact)	−0.074	0.229	−0.322	0.747	0.929	[0.593, 1.455]	ns
Homozygous (vs intact)	+0.303	0.097	+3.125	0.0018	1.353	[1.119, 1.636]	★★ p<0.01
Sex (male vs female)	+0.166	0.099	+1.679	0.093	1.180	[0.973, 1.433]	ns

C-INDEX	HOMOZYGOUS HR	HOMOZYGOUS P	N (MODEL)	EVENTS	REFERENCE
0.535	1.353	0.0018	555	455	intact (n=335)

- CDKN2A homozygous loss:
  - 35.3% higher death risk
- CDKN2A heterozygous loss:
  - not significant
- Sex:
  - not significant
- C-index:
  - the model performance is similar to random ranking

Conclusion

- CDKN2A homozygous loss has a biological effect on survival (p-value = 0.002)
  - frequent modification in GBM
- CDKN2A homozygous loss is quite frequent (33%) increases the risk of death by 35%
- Heterozygous loss was present in low n in our dataset, but no apparent affect was detected
- Concordance index (0.53) suggests that CDKN2A alone is not useful as a biomarker
  - include more variables to gain a better prognosis
- Outlook: What effect does the deletion of CDKN2B have?

Sources

- progenetix: https://progenetix.org/search/?
- ensembl: https://www.ensembl.org/Homo_sapiens/Gene/Summary?g=ENSG00000147889;r=9:21967752-21995301
- UniProt: https://www.uniprot.org/uniprotkb/P42771/entry
- Intergrative Genomics Viewer: https://igv.org/
- Królikowska, K., Błaszczak, K., Ławicki, S., Zajkowska, M., & Gudowska-Sawczuk, M. (2025). Glioblastoma-A Contemporary Overview of Epidemiology, Classification, Pathogenesis, Diagnosis, and Treatment: A Review Article. *International journal of molecular sciences*, *26*(24), 12162. https://doi.org/10.3390/ijms262412162
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